A torsion-free path violates the alternating tree sum

Conjecture 00000004228

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Convention. We use integral cellular homology and the usual cellular spanning-tree counts: a zero-dimensional tree is a single vertex; a one-dimensional spanning tree is a connected acyclic spanning subgraph. For graphs, unweighted and torsion-weighted spanning-tree counts agree. The original conjecture supplies no alternative definition of tree counts.

Let X be the path 0–1–2, with oriented edges $0 \to 1$ and $1 \to 2$. Its complete cellular chain complex is

$$0 \longrightarrow C_1 = \mathbb{Z}^2 \xrightarrow{\partial} C_0 = \mathbb{Z}^3 \longrightarrow 0, \qquad \partial(a, b) = (-a, a - b, b).$$

Write $s(x, y, z) = x + y + z$ for augmentation.

Theorem 1. *All integral homology groups of X are torsion-free, but $\chi(X) \neq \tau_0(X) - \tau_1(X)$.*

Proof. The equation $\partial(a, b) = 0$ forces $a = b = 0$, so $H_1(X) = 0$. Moreover, $s\partial = 0$ and every (x, y, z) with $x + y + z = 0$ satisfies

$$\partial(-x, z) = (x, y, z).$$

Consequently $\operatorname{im} \partial = \ker s$ and $H_0(X) \cong \mathbb{Z}$. All higher chain groups and homology groups are zero. This proves the absence of torsion in every degree.

There are three vertices and two edges, so $\chi(X) = 3 - 2 = 1$. There are three zero-dimensional trees, one for each vertex. A spanning subgraph is connected only if it contains both path edges, and that subgraph is acyclic. Thus $\tau_0(X) = 3$ and $\tau_1(X) = 1$, giving

$$\tau_0(X) - \tau_1(X) = 3 - 1 = 2 \neq 1 = \chi(X).$$

This refutes the torsion-free implication; the unspecified correction term in the presence of torsion is not needed. □

Lean correspondence. `EulerTrees.lean` defines the integral chain groups and proves $\partial^2 = 0$, the kernel, and the image assertions. The predicate `HomologyTorsionFree` states the exact quotient criterion

$$q > 0, \quad z \in Z_n, \quad qz \in B_n \quad \Longrightarrow \quad z \in B_n.$$

Thus `homology_torsion_free` proves torsion-freeness directly, without assuming homology ranks or omitting higher degrees.

For edge masks, `IsSpanningTree` requires integral connectivity and a zero cycle kernel. `spanning_tree_iff` proves that only the full mask is a tree. The mask list is exhaustive and duplicate-free; its filter is proved equivalent to that predicate. The final theorem combines all-degree torsion-freeness and the unequal Euler and tree counts.

Scope and verification. This is a counterexample under the stated standard convention, not a result about an unspecified alternative formula. Lean 4.19.0, core library only; the audit uses only `propext`, `Classical.choice`, and `Quot.sound`. Competition acceptance has not been claimed.

Source. The official bilingual problem is included as `conjecture.md`. For cellular trees and torsion weights, see Duval, Klivans and Martin, Simplicial and Cellular Trees.